Beyond the coding genome in metastatic cutaneous squamous cell carcinoma

Amarinder S. Thind [1,3*], Bruce Ashford [1,2,3,10*], Dario Strbenac[4], Ruta Gupta[5], Jonathan Clark[6,7], N. Gopalakrishna Iyer [8], Elahe Minaei [3,9,10], Jay Perry [3,9,10] Jenny Lee [7] and Marie Ranson [3,9,10].

1 School of Medicine, University of Wollongong, Wollongong, Australia, 2 Illawarra Shoalhaven Local Health District Wollongong, Australia, 3 Illawarra Health and Medical Research Institute, Wollongong, Australia, 4 University of Sydney, Sydney Australia, 5 Anatomical Pathology, Royal Prince Alfred Hospital, Sydney, Australia, 6 Sydney Head and Neck Cancer Institute, Sydney, Australia, 7 Chris O'Brien Lifehouse Royal Prince Alfred Hospital, Sydney, Australia, 8 National Cancer Center Singapore, 9 School of Chemistry and Molecular Bioscience, University of Wollongong, Wollongong, Australia, 10 Centre for Oncology, Education and Research Translation, NSW, Australia.

Abstract

There is limited published data exploring whole genome sequencing (WGS) in metastatic cutaneous squamous cell carcinoma (cSCC). In this study, we used WGS on matched tumor and blood DNA to detect somatic variants from 25 patients with regional metastases of head and neck cSCC. Functional level analysis of structural, copy number and short variants was performed utilising a combination of bioinformatic tools to interrogate their clinical impacts on metastasis across the cohort. In the non-coding genome, 3'UTR regions of EVC (48%), PPP1R1A (48%) and LUM (16%) were significantly functionally altered (Q-value < 0.05). Further, significant functional alterations are observed in the tumor suppressing lncRNA LINC01003 (68% of specimens, Q-value: 0.0158). In addition, significant recurrent copy number loss in tumor suppressor genes KANSL1 and PTPRD and gain in CALR, CCND1 and FGF3 was observed for coding regions. Significantly mutated gene analysis predicted TP53, CDKN2A, ZNF750 as potential drivers of the metastasis cSCC.

Objectives

To identify non-coding genomic mutational drivers and their functional impact in metastatic cSCC in the clinical context.

Materials and Methods

- Metastatic cSCC patients whole blood samples and, sections of fresh tumor from nodal metastases were collected during surgery.
- Only samples with histopathologically assessed > 30% cellularity were sequenced. Only 25/32 samples passed the Quality control. The remaining 7 samples had insufficient tumor content or had an extreme GC Bias.
- Somatic calling performed with Mutech2 pipeline. Further somatic variants are also adjusted for tumor purity using PURPLE (Priestley et al., 2019)
- Variants effect prediction and annotations completed using OpenCravat platform (open-cravat.readthedocs.io). Functional impact of both coding and non-coding variants was predicted using the ONCOdriveFML tool (Mularoni et al., 2016).
- Using PURPLE, we derived copy number alterations analysis on 25 metastatic specimens.
- LINX was employed for structural variant analysis (Cameron et al., 2019). We combined copy number variation, short coding variants and other key regulatory variants based coding and noncoding genes for an enrichment analysis.

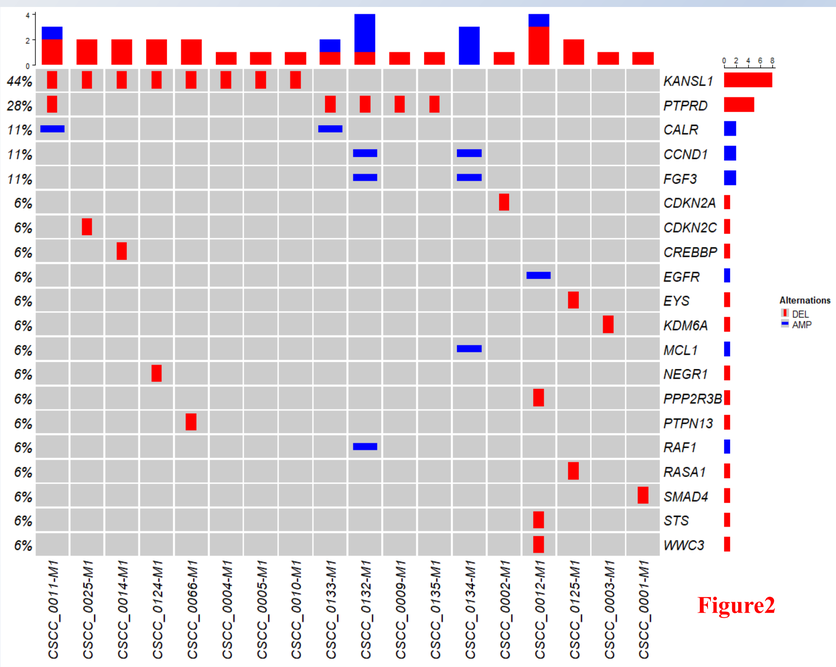
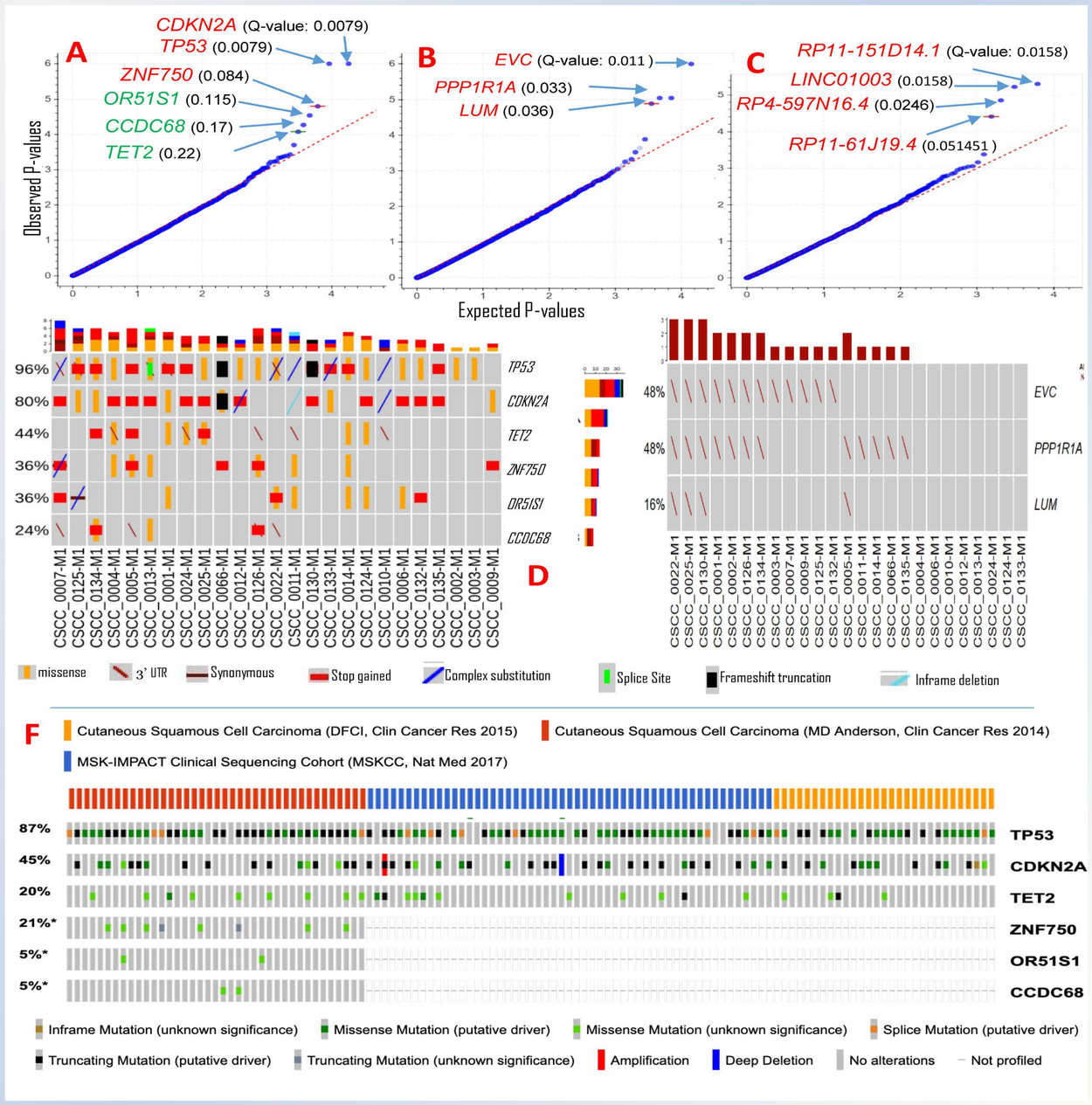


Figure2: Recurrent copy number gain and loss for individual genes,. These driver genes were derived by the use of purple-3.1.

Copy Number Alterations

The most frequently deleted genes were KANSL1 and PTPRD. KANSL1 deletion observed in 44% of samples. Deletion of PTPRD was seen in 28% of samples.

Results



Mutations in non-coding regulatory regions

Within the 3'UTR region of *EVC, LUM* and *PPP1R1A*, recurrent specific non-coding SNVs showed significant higher observed functional impact than the predicted functional impact measured by randomized mutations with the same number of variants within the specified co-ordinates of the 3'UTR regions (Figure 1b).

Figure 1 (a –c) The most significantly altered (a) coding region (genes), (b) 3' UTR regions of the genes, and (c) lncRNA regions. Plots showing the frequency of observed mutations with respect to the expected frequency of the mutations in the corresponding regions (OncodriveFML (2.2.0)). (d) Detailed sample-level information of the SNVs in the top altered genes (refer to panel a). (e) Variants of significantly altered 3'UTR regions. (f). Oncplot retrieved from cBioportal to compare the variants from panel (d) with other metastatic cSCC studies.

lncRNA LINC01003, is amongst the most mutated lncRNA in our cohort (71% of samples). Other recurrently varied lncRNA in our cohort were RP11-151D14.1, RP4-597N16.4 and RP11-61J19.4 (Figure 1c).

Coding short variants

Significantly mutated gene analysis predicted TP53, CDKN2A, ZNF750 as potential drivers of the metastasis cSCC. Over 80% of SNV mutations were C>T, consistent with the dominant effect of UV radiation on pyrimidine bases.(Figure 1A, 1D and Figure 3).

Structural variation and gene fusions

Fragmentation and rearrangement of genomic regions was seen across many samples. Only a single coding-coding gene fusion was observed in our cohort – this was between STRN and DLG2 in sample CSCC_0009 (STRN: Exon 1 ENST00000263918 DLG2 Exon 7 ENST00000376104).

Statistical Challenges for significance detection in Promotor/5'UTR regions of cSCC:

This phenomenon such as decreased level of nucleotide excision repair (NER) or high mutation rate at active transcription factor binding sites (TFBS) due to E26 transformation-specific (ETS) TFs (which induced cyclobutane pyrimidine dimers (CPDs) hotspots) leads to many false-positive mutational driver detections from promotor/5' UTR regions.

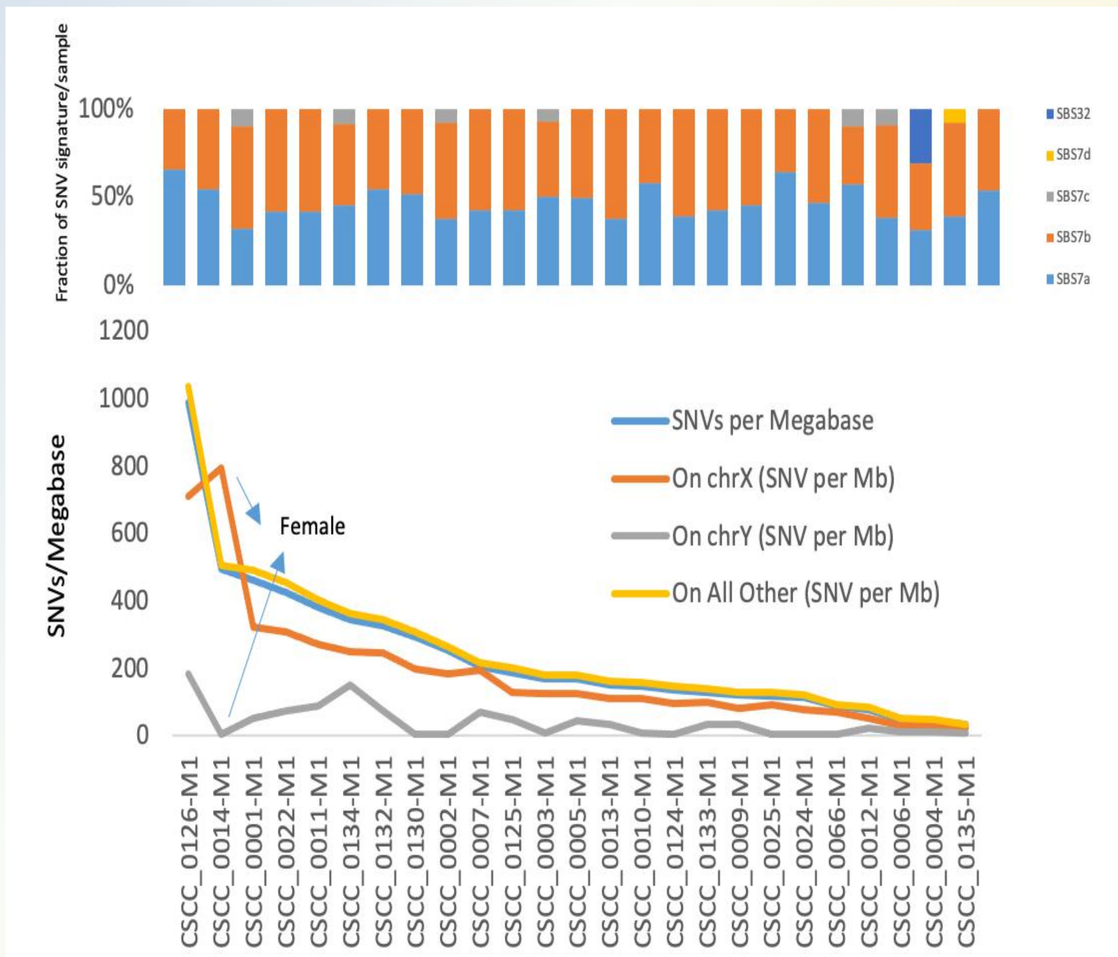


Figure 3: (A) Various mutational signatures based on Cosmic3 are shown. (b) Tumor mutational burden per Megabase (sorted in decreasing order)

The average TMB for SNV across the 25 cases was 238.7 mutations per megabase (range 32.52to 995.66 mutations/Mb). As we have previously reported, a mutation density across the genome 171 times higher for non-coding variants compared to coding variants [Muller et. Al 2019] . SNV signature analysis of our 25 samples shows SBS signatures 7a and 7b are dominant (Figure: 3) Substantial representation of SBS7c is also seen. SBS32 and SBS7d are observed in one sample .

Conclusions

We have previously shown that non-coding variants dominate the genome of metastatic cutaneous SCC and highlighted the need for whole genome sequencing (WGS) to fully elucidate the mutational landscape of this disease [Muller et. al. 2019]. Herein, we examined the patterns and frequency of non-coding variants in key regulatory regions, including the untranslated regions (UTR) and long non-coding RNA and describe their potential functional consequences.

References

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